

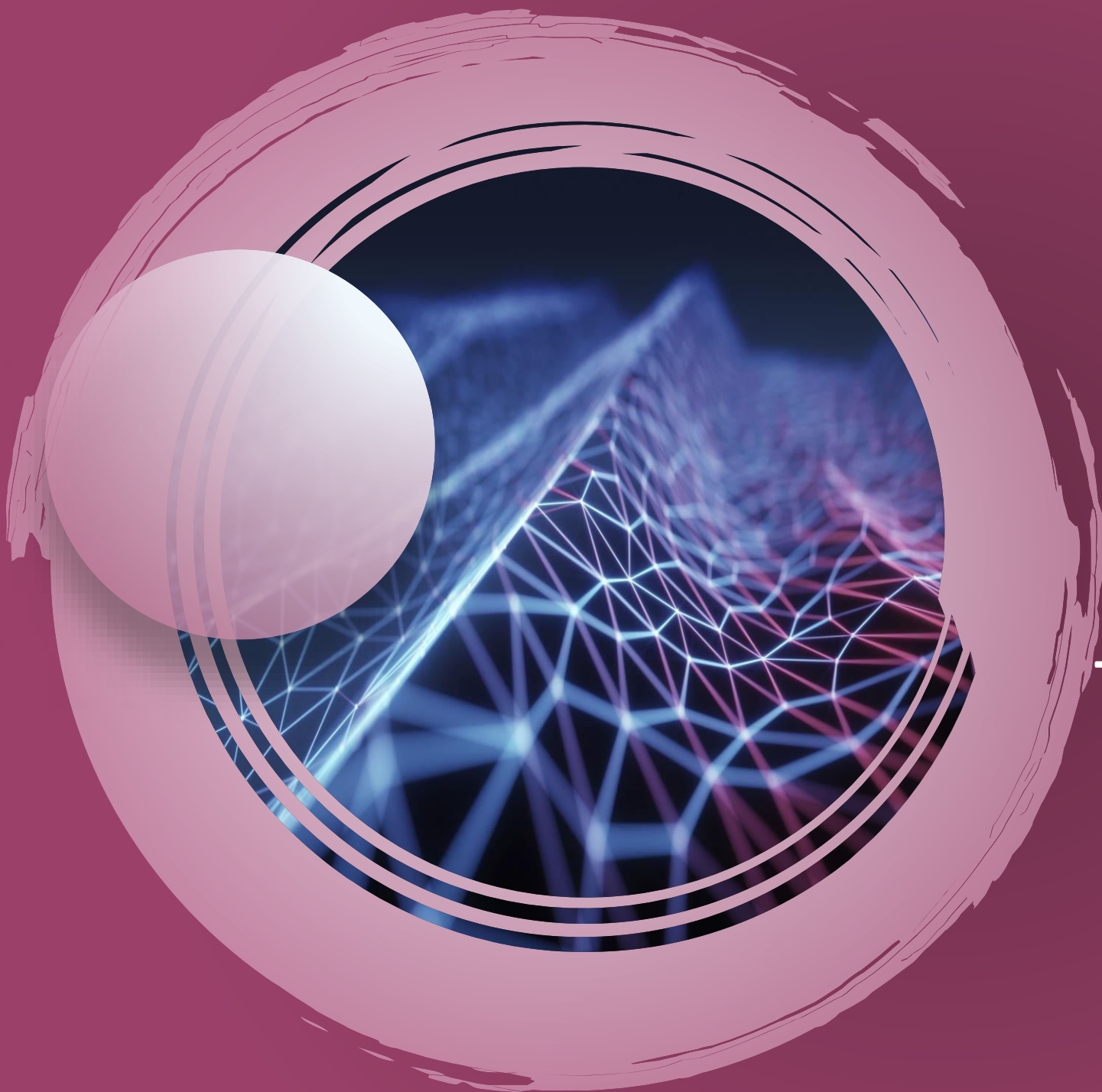
WELCOME

Tajwar Mehmood

Office: A209G

tajwar.mehmood@nu.edu.pk

<https://tmehmood1225.wixsite.com/tajwar>

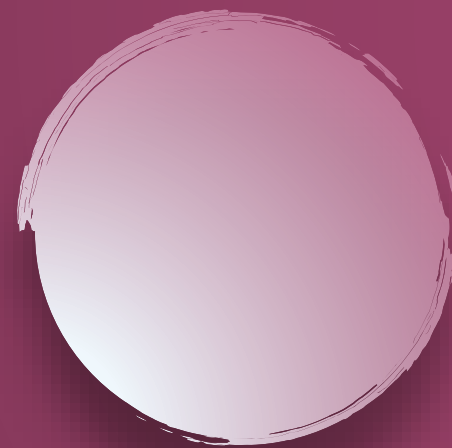


THEORY OF AUTOMATA

WEEK 2

Contents

- Recursive Definition
 - Examples
- Arithmetic Expression
 - Recursive Definition of AE
- Theorems
- Propositional Calculus
 - Well Formed Formula



RECURSIVE DEFINITIONS

Recursive definition of languages

The following three steps are used in recursive definition

1. Some basic words are specified in the language.
2. Rules for constructing more words are defined in the language.
3. No strings except those constructed in above, are allowed to be in the language.

RECURSIVE DEFINITIONS

EXAMPLE

1. EVEN
2. PALINDROME
3. LANGUAGE $\{A^nB^n\}$

Defining language of **EVEN**

Step 1:

2 is in **EVEN**.

Step 2:

- a. If x is in **EVEN** then $x+2$ and $x-2$ are also in **EVEN**.
- b. If x and y are in **EVEN** then so are $x+y$, $x-y$ and $x*y$.

Step 3:

No strings except those constructed in above, are allowed to be in **EVEN**.

Defining language of PALINDROME

- Defining the language PALINDROME, defined over $\Sigma = \{a,b\}$

Step 1:

λ , a and b are in PALINDROME

Step 2:

if x is palindrome then axa , bxb , xx are also be palindrome,

Step 3:

No strings except those constructed in above, are allowed to be in palindrome

Defining language $\{a^n b^n\}$

- Defining the language $\{a^n b^n\}$, $n=1,2,3,\dots$, of strings defined over $\Sigma=\{a,b\}$

Step 1:

ab is in $\{a^n b^n\}$

Step 2:

if x is in $\{a^n b^n\}$, then axb is in $\{a^n b^n\}$

Step 3:

No strings except those constructed in above, are allowed to be in $\{a^n b^n\}$

Defining language L, of strings ending in a

- Defining the language L, of strings ending in a , defined over $\Sigma=\{a,b\}$

Step 1:

a is in L

Step 2:

(s)a is also in L, where s belongs to Σ^*

Step 3:

No strings except those constructed in above, are allowed to be in L

Defining language

- Defining the language L , of strings beginning and ending in same letters, defined over $\Sigma = \{a, b\}$

Step 1:

a and b are in L

Step 2:

$(a)s(a)$ and $(b)s(b)$ are also in L , where s belongs to Σ^*

Step 3:

No strings except those constructed in above, are allowed to be in L

Defining language

- Defining the language L , of strings beginning and ending in a letters, defined over $\Sigma = \{a, b\}$

Step 1:

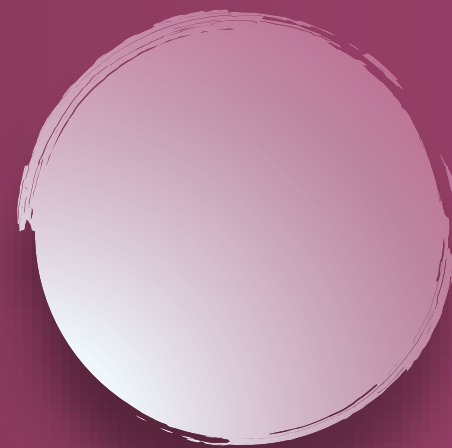
a are in L

Step 2:

$(a)s(a)$ are also in L , where s belongs to Σ^*

Step 3:

No strings except those constructed in above, are allowed to be in L



ARITHMETIC EXPRESSIONS

Arithmetic Expressions

- Suppose we ask ourselves what constitutes a valid arithmetic expression, or AE for short.
- The alphabet for this language is
- $\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, +, -, *, /, (,)\}$

Arithmetic Expression AE

- Obviously, the following expressions are not valid:

$(3 + 5) + 6)$ $2(/8 + 9)$ $(3 + (4-)8)$

- The first contains unbalanced parentheses; the second contains the forbidden substring (/; the third contains the forbidden substring -).
- Are there more rules? The substrings $//$ and $*/$ are also forbidden.
- Are there still more?
- The most natural way of defining a valid AE is by using a **recursive definition**, rather than a long list of forbidden substrings.

Recursive Definition of AE

- **Rule 1:** Any number (positive, negative, or zero) is in AE.
- **Rule 2:** If x is in AE, then so are
 - (i) x
 - (ii) $-x$ (provided that x does not already start with a minus sign)
- **Rule 3:** If x and y are in AE, then so are
 - (i) $x + y$ (if the first symbol in y is not $+$ or $-$)
 - (ii) $x - y$ (if the first symbol in y is not $+$ or $-$)
 - (iii) $x * y$
 - (iv) x / y
 - (v) $x ** y$ (our notation for exponentiation)

Recursive Definition of AE

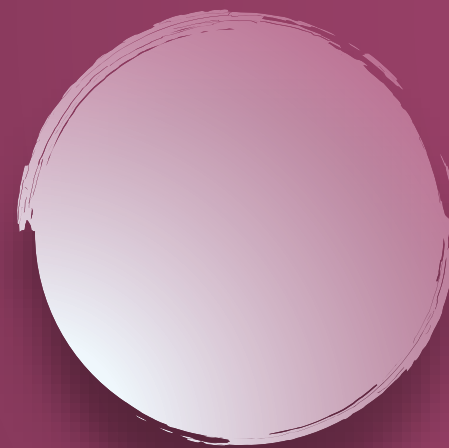
- The above definition is the most natural, because it is the method we use to recognize valid arithmetic expressions in real life.
- For instance, we wish to determine if the following expression is valid:

$$(2 + 4) * (7 * (9 - 3)/4)/4 * (2 + 8) - 1$$

- We do not really scan over the string, looking for forbidden substrings or count the parentheses.
- We actually imagine the expression in our mind broken down into components:
 - Is $(2 + 4)$ OK? Yes
 - Is $(9 - 3)$ OK? Yes
 - Is $7 * (9 - 3)/4$ OK? Yes, and so on.

Recursive Definition of AE

- Note that the recursive definition of the set AE gives us the possibility of writing $8/4/2$, which is ambiguous, because it could mean $8/(4/2) = 4$ or $(8/4)/2 = 1$.
- However, the ambiguity of $8/4/2$ is a problem of *meaning*. There is no doubt that this string is a word in AE, only doubt about what it means.
- By applying Rule 2, we could always put enough parentheses to avoid such a confusion.
- The recursive definition of the set AE is useful for proving many theorems about arithmetic expressions, as we shall see in the next few slides.



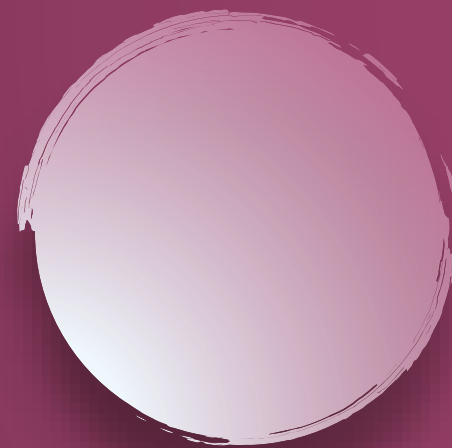
THEOREM

Theorem

- **An arithmetic expression cannot contain the character \$.**
- *Proof*
- This character is not part of any number, so it cannot be introduced into an AE *by Rule 1*.
- If the character string x does not contain the character \$, then neither do the string (x) and $-x$. So, the character \$ cannot be introduced into an AE *by Rule 2*.
- If neither x nor y contains the character \$, then neither do any of the expressions defined in *Rule 3*.
- Therefore, the character \$ can never get into an AE.

Theorem 3 & 4

- **No arithmetic expression can begin or end with the symbol $/$.**
- **Proof?**
- **No arithmetic expression can contain the substring $//$.**
- **Proof?**



PROPOSITIONAL CALCULUS

Propositional Calculus

- Propositional calculus (or sentential calculus) is a branch of symbolic logic that we shall be interested in.
- The version we define here uses only negation ($\neg$) and implication ($\rightarrow$), together with the phrase variables.
- The alphabet for this language is
 - $\Sigma = \{\neg, \rightarrow, (,), a, b, c, d, \dots\}$
- A valid expression in this language is called WFF (**well-form formula**).

Propositional Calculus

- The rules for forming WFFs are:
- **Rule 1:** Any single Latin letter is a WFF, for instance a , b , c , ...
- **Rule 2:** If p is a WFF, then so are (p) and $\neg p$.
- **Rule 3:** If p and q are WFFs, then so is $p \rightarrow q$.
- Can you show that $p \rightarrow ((p \rightarrow p) \rightarrow q)$ is a WFF?
- Can you show that the following are NOT WFFs?
 - $p \rightarrow$
 - $\rightarrow p$
 - $p) \rightarrow p($

Also in mathematics we often see the following definition of factorial:

Rule 1 $0! = 1$

Rule 2 $(n + 1)! = (n + 1)(n!)$

The reason that these definitions are called “recursive” is that one of the rules used to define the set mentions the set itself. We define EVEN in terms